

Where you stand in Chapter 2

WRITE THIS CODE ON YOUR PAGE

D1

10 questions, chosen to split what is still uncertain about your skills

HOW TO WRITE YOUR WORK

- Write the code **D1** at the top of every page you photograph.
- Start each problem with its number. One step per line.
- Cross mistakes out with a single line; do not erase.
- Box your final answer.
- If you have not seen a topic yet, write the number and *skip*. Skipping is information too.

1. Find the slope of the line through the points $(2, f(2))$ and $(3, f(3))$ on the graph of $f(x) = x^2 - 3$.
2. Rationalize the denominator: $\frac{1}{\sqrt{x} - 4}$.
3. Simplify for $x \neq 0, 4$: $\frac{\frac{1}{x} - \frac{1}{4}}{x - 4}$.
4. For which values of x is $f(x) = \frac{x - 4}{x^2 + 11x + 30}$ undefined?
5. Solve $|x + 3| < 1$. Give the answer in interval notation.

6. Simplify $\frac{\sin^2(x)}{1 - \cos(x)}$ (where defined).
7. Let $f(x) = x^2 + 1$. Find the slope of the secant line through $P(1, f(1))$ and $Q(2, f(2))$.
8. A ball's height is $s(t) = -16t^2 + 48t$ feet after t seconds. Find its average velocity over $\left[2, \frac{5}{2}\right]$.
9. Estimate the area between the x -axis and $f(x) = x^2$ over $[0, 2]$ using 2 rectangles of equal width with heights at the right endpoints.
10. Given $\lim_{x \rightarrow 2} f(x) = -4$ and $\lim_{x \rightarrow 2} g(x) = 3$, evaluate $\lim_{x \rightarrow 2} \frac{4f(x) - 1g(x)}{f(x)g(x)}$.